13 Implement the Car Price Prediction Model using Python
Code:

```
import matplotlib.pyplot as plt
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Student data: [Study Hours, Attendance %]
X = [
    [1, 40], [2, 50], [2, 60], [3, 65],
    [4, 70], [5, 75], [6, 80], [7, 90],
    [3, 55], [5, 85]
]

# 0 = Fail, 1 = Pass
y = [0, 0, 0, 0, 1, 1, 1, 1, 0, 1]

# Train Random Forest model
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X, y)

# Predict results
y_pred = model.predict(X)

print("Random Forest Classification")
print("Training Accuracy:",
      round(accuracy_score(y, y_pred) * 100, 2), "%")

# Predict for a new student
student = [[4, 72]]
prediction = model.predict(student)

result = "Pass" if prediction[0] == 1 else "Fail"
print("Study Hours:", student[0][0])
print("Attendance:", student[0][1], "%")
print("Predicted Result:", result)

# Plot feature importance
features = ["Study Hours", "Attendance"]
importance = model.feature_importances_

plt.bar(features, importance)
plt.xlabel("Features")
plt.ylabel("Importance")
```

```
plt.title("Random Forest - Feature Importance")
plt.grid(axis="y")
plt.show()
```

Output:

```
Random Forest Classification
Training Accuracy: 100.0 %
Study Hours: 4
Attendance: 72 %
Predicted Result: Pass
```

